Experiment 02

Title

Data Preprocessing and Exploratory Data Analysis (EDA)

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AIM

To perform Exploratory Data Analysis (EDA), clean and preprocess data, handle missing values, scale features, perform feature engineering, and apply dimensionality reduction techniques to prepare a model-ready dataset.

THEORY

Data preprocessing is an essential step in machine learning. It includes:

- **EDA**: Understanding distributions, correlations, and patterns in data
- **Missing Value Handling**: Using imputation techniques
- **Outlier Detection**: Identifying abnormal values using IQR
- **Feature Scaling**: Standardizing features using Z-score normalization
- **Feature Engineering**: Creating new meaningful features
- **Imbalanced Data Handling**: Using resampling techniques
- **Dimensionality Reduction**: Reducing features using PCA

Commands/Code Used

```python
# Full Experiment 2 code (EDA + Preprocessing)

from sklearn.datasets import load_iris
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

iris = load_iris()
df = pd.DataFrame(iris.data, columns=iris.feature_names)
df["target"] = iris.target

# Correlation heatmap
```

```
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(), annot=True, cmap="coolwarm")
plt.savefig("eda_correlation.png")

# Distribution plots
df.hist(figsize=(10,8))
plt.savefig("eda_distributions.png")

# PCA
X = df.iloc[:, :-1]
y = df["target"]

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

plt.figure(figsize=(8,6))
plt.scatter(X_pca[:,0], X_pca[:,1], c=y, cmap="viridis")
plt.colorbar()
plt.title("PCA: Dimensionality Reduction")
plt.savefig("pca_reduction.png")
```

Results

1. Distribution Analysis

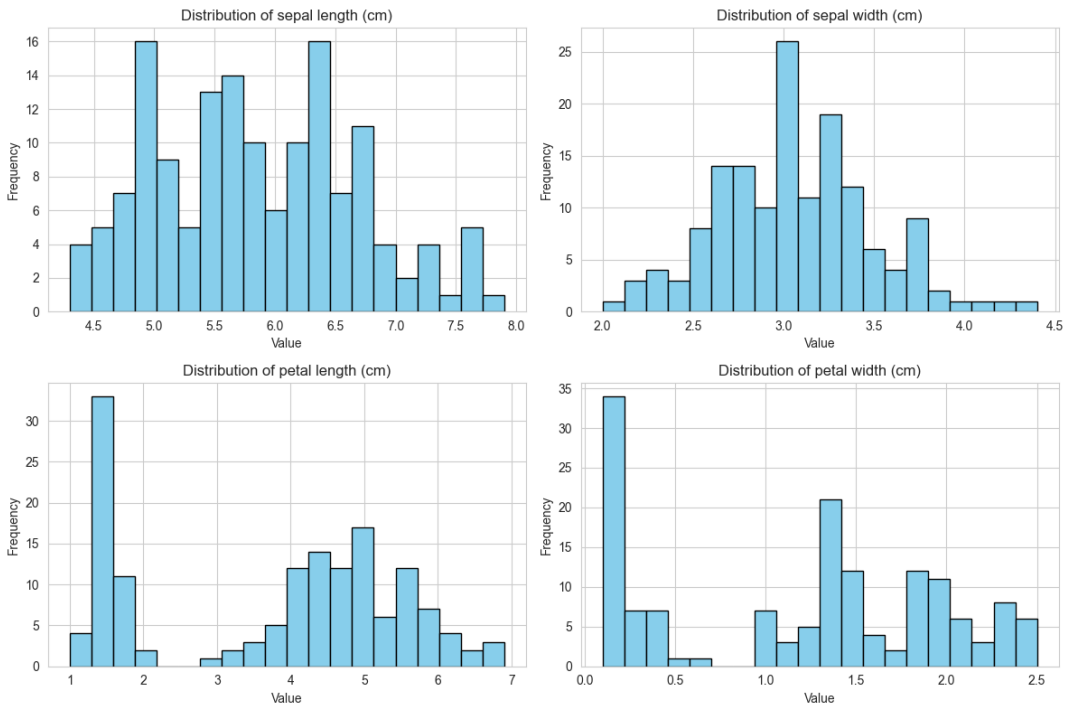

Figure 1: Distribution of features in Iris dataset

2. Correlation Heatmap

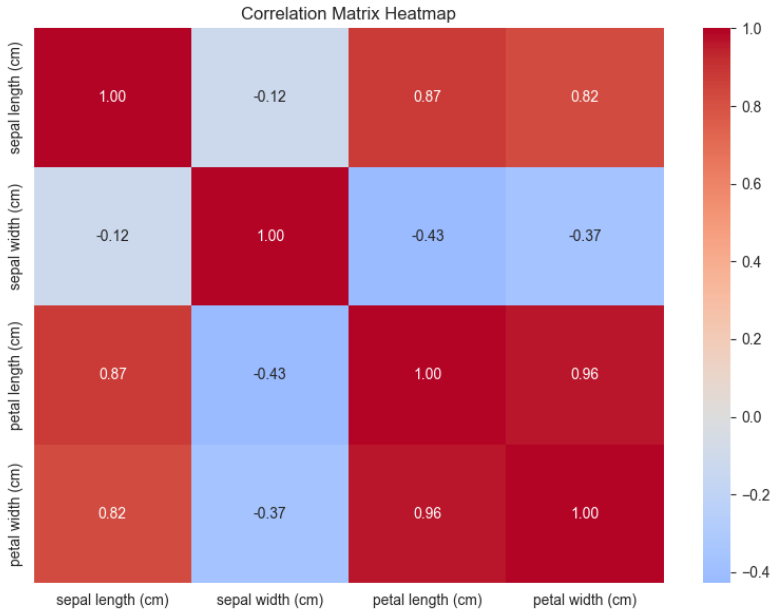

Figure 2: Correlation matrix showing relationships between features

3. PCA Visualization

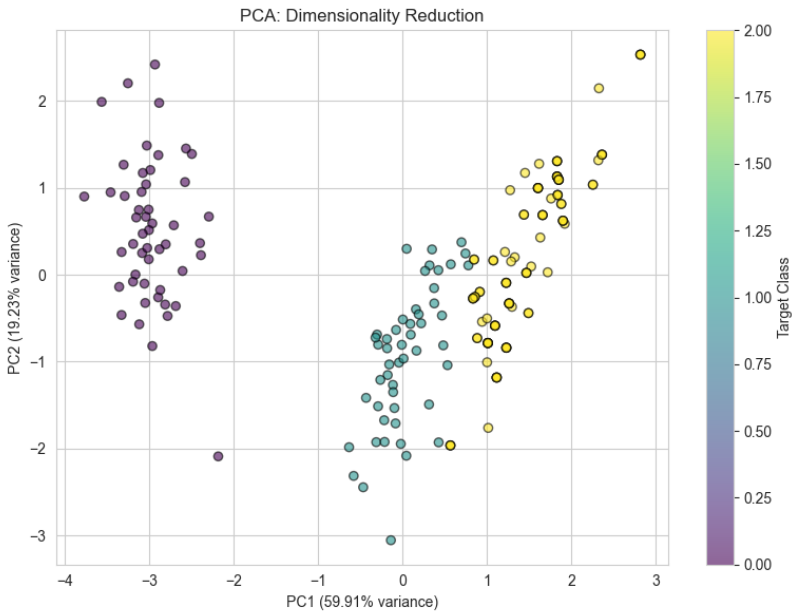

Figure 3: PCA-based dimensionality reduction

Conclusion

In this experiment, we successfully performed data preprocessing and exploratory data analysis. We analyzed feature distributions, visualized correlations, handled data transformation, and applied PCA for dimensionality reduction. These steps are crucial for improving model performance and preparing data for machine learning algorithms.